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METHOD FOR OPERATING POWER CONVERTER, SEMICONDUCTOR CHIP AND POWER CONVERTER

Abstract

A power converter includes a transformer, a clamp capacitor, a main switch, a clamp switch, and a control circuit. The transformer has a first primary coil and a secondary coil. The clamp capacitor has a first terminal coupled to a first terminal of the first primary coil. The main switch is coupled to a second terminal of the first primary coil. The clamp switch is coupled between a second terminal of the clamp capacitor and the main switch. The main switch, the clamp switch, and the first primary coil intersect at a node. The control circuit is configured to: periodically turn on the main switch; turn on the clamp switch before the main switch to generate a reverse current; turn off the clamp switch before the main switch to discharge an equivalent capacitor; and adjust an operation of the clamp switch based on a monitored voltage at the node.

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Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/193,593, filed Mar. 30, 2023, and entitled "METHOD FOR OPERATING POWER CONVERTER, SEMICONDUCTOR CHIP AND POWER CONVERTER, which claims priority to Chinese patent application No. 202211476002.9, filed on Nov. 23, 2022 and entitled "METHOD FOR OPERATING POWER CONVERTER, SEMICONDUCTOR CHIP AND POWER CONVERTER", the disclosures of the applications are hereby incorporated by reference in their entirety for all purposes.

TECHNICAL FIELD
[0002] The present invention relates to a power converter, a semiconductor chip, and a method for operating the power converter, in particular to a power converter capable of reducing switching loss.
BACKGROUND
[0003] Flyback power converters are often used in a variety of electronic products because of their simple architecture and ability to provide electrical isolation through a transformer therein. However, when switching a switch of the primary coil, the conventional flyback power converter does not discharge the parasitic capacitor on the terminal of the switch, but turns on the switch in the form of a hard switching, so the switching loss is serious. Therefore, how to design a more efficient power converter has become an urgent problem to be solved.
SUMMARY
[0004] Embodiments of the present disclosure relate to a power converter. The power converter includes a transformer, a clamp capacitor, a main switch, a clamp switch, and a control circuit. The transformer includes a first primary coil and a secondary coil, where a first terminal of the first primary coil is used for receiving an input voltage, and the secondary coil is used for generating an output voltage. The clamp capacitor has a first terminal coupled to the first terminal of the first primary coil. The main switch is connected in series between a second terminal of the first primary coil and a ground terminal. The clamp switch has a first terminal coupled to a second terminal of the clamp capacitor, and a second terminal coupled to a first terminal of the main switch, the clamp switch is connected in series between the clamp capacitor and the main switch, and the second terminal of the clamp switch is further coupled to the second terminal of the first primary coil, so that the second terminal of the clamp switch, the first terminal of the main switch and the second terminal of the first primary coil intersect at a common node. The control circuit includes a voltage detector and a control signal generator. The voltage detector is used for comparing a node voltage of the common node with a first threshold voltage. The control signal generator is used for periodically: turning on the main switch; turning on the clamp switch to generate a reverse current into the second terminal of the primary coil before the main switch is turned on; turning off the clamp switch before the main switch is turned on to discharge an equivalent capacitor on the common node through the reverse current after the clamp switch is turned on for a period; and determining a length of time for which the clamp switch is turned on before the main switch is

turned on next time according to a magnitude relationship between the node voltage and the first threshold voltage detected by the voltage detector after the clamp switch is turned off and at a predetermined time point before the main switch is turned on, thereby controlling the voltage of the common node before the main switch is turned on next time to suppress a switching loss of the main switch.

[0005] Another embodiment of the present disclosure relates to a chip including the control circuit in the power converter.

[0006] Another embodiment of the present disclosure relates to a method for operating a power converter. The power converter includes a transformer, a main switch, a clamp capacitor and a clamp switch, the transformer includes a primary coil and a secondary coil, the main switch is connected in series between a second terminal of the primary coil and a ground terminal, a first terminal of the clamp switch is coupled to a second terminal of the clamp capacitor, a second terminal of the clamp switch is coupled to the first terminal of the main switch and the second terminal of the primary coil, and the second terminal of the clamp switch, the first terminal of the main switch and the second terminal of the primary coil intersect at a common node. The method includes periodically: turning on the main switch to charge the primary coil with an input voltage; turning on the clamp switch to generate a reverse current into the second terminal of the primary coil before the main switch is turned on; turning off the clamp switch before the main switch is turned on to discharge the equivalent capacitor on the common node through the reverse current after the clamp switch is turned on for a period; detecting a magnitude relationship between a node voltage of the common node and a first threshold voltage at a predetermined time point after the clamp switch is turned off and before the main switch is turned on; and determining a length of time for which the clamp switch is turned on before the main switch is turned on next time according to a magnitude relationship between the node voltage and the first threshold voltage detected by the voltage detector after the clamp switch is turned off and at a predetermined time point before the main switch is turned on, thereby controlling the voltage of the common node before the main switch is turned on next time to suppress a switching loss of the main switch.

[0007] Since the length of time for which the clamp switch is turned on next time can be determined by the power converter of the present disclosure according to the amplitude of the node voltage of the common node before the main switch is turned on each time, the voltage of the common node can be pulled down to a predetermined range in different circuits and environments, thereby effectively suppressing a switching loss of the main switch.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] Aspects of several embodiments of the disclosure may be best understood from the following detailed description when read with the accompanying drawings. It will be noted that the various structures may not be drawn to scale. In fact, the dimensions of the various structures may be arbitrarily expanded or reduced for clarity of discussion.

[0009] FIG. **1** is a schematic diagram of a power converter according to an embodiment of the present disclosure;

[0010] FIG. **2** is a flow chart of a method for operating the power converter of FIG. **1** according to an embodiment of the present disclosure;

[0011] FIG. **3** is an operation timing diagram of the power converter of FIG. **1** according to the method for FIG. **2**;

[0012] FIG. **4** is an operation timing diagram of the power converter of FIG. **1** according to another embodiment;

[0013] FIG. **5** is an operation timing diagram of the power converter of FIG. **1** according to another

embodiment;
[0014] FIG. **6** is a schematic diagram of a power converter according to another embodiment of the present disclosure;
[0015] FIG. **7** is a schematic diagram of a power converter according to another embodiment of the present disclosure;
[0016] FIG. **8** is a schematic diagram of a power converter according to another embodiment of the present disclosure.
[0017] The same reference numbers are used throughout the drawings and the detailed description to refer to the same or like parts. Several embodiments of the present disclosure will be readily understood from the following detailed description taken in conjunction with the accompanying drawings.
DETAILED DESCRIPTION OF THE EMBODIMENTS
[0018] The following disclosure provides many different embodiments or examples for implementing different features of the provided subject matter. Specific examples of components and configurations are described below. These are, of course, merely examples and are not intended to be limiting. In the present disclosure, references to forming a first feature over or on a second feature may include embodiments that form the first and second features into direct contact, and may also include embodiments that may form additional features between the first and second features such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the sake of simplicity and clarity and does not dictate a relationship between the various embodiments and/or configurations discussed.
[0019] Embodiments of the present disclosure are discussed in detail below. However, the present disclosure provides many applicable concepts that can be embodied in a wide variety of specific contexts. The embodiments discussed are merely illustrative and do not limit the scope of the disclosure.
[0020] The present disclosure provides a power converter, a semiconductor chip, and a method for operating the power converter. The disclosed power converter includes a transformer, a clamp capacitor, a main switch, a clamp switch, and a control circuit. The first terminal of the clamp capacitor is coupled to the first terminal of the primary coil of the transformer, and the main switch is connected in series between the second terminal of the primary coil and a ground terminal. The clamp switch is connected in series between the second terminal of the clamp capacitor and the first terminal of the main switch, and the second terminal of the clamp switch, the first terminal of the main switch and the second terminal of the primary coil intersect at a common node. The control circuit periodically turns on the main switch and turns on the clamp switch for a period before turning on the main switch to generate a reverse current. After the clamp switch is turned on and before the main switch is turned on, the control circuit turns off the clamp switch to discharge an equivalent capacitor on the common node, and determines the length of time for which the clamp switch is turned on next time according to the relationship between the voltage of the common node and the threshold voltage before the main switch is turned on. Since the length of time for which the clamp switch is turned on next time can be determined by the power converter of the present disclosure according to the amplitude of the node voltage of the common node before the main switch is turned on each time, the voltage of the common node can be pulled down to a predetermined range in different circuits and environments, thereby effectively suppressing a switching loss of the main switch.
[0021] FIG. **1** is a schematic diagram of a power converter according to an embodiment of the present disclosure. As shown in FIG. **1**, a power converter **100** may include a transformer **110**, a clamp capacitor **120**, a main switch **130**, a clamp switch **140**, and a control circuit **150**. In this embodiment, the power converter may be, for example, an active clamp flyback power converter.
[0022] The transformer **110** includes a primary coil **112** and a secondary coil **114**. A first terminal

of the primary coil **112** may receive an input voltage Vin, and the secondary coil **114** may induce and output an output voltage V.sub.out according to a change in current on the primary coil **112**. The clamp capacitor **120** has a first terminal and a second terminal, and the first terminal of the clamp capacitor **120** may be coupled to the first terminal of the primary coil **112**. The main switch **130** is connected in series between the second terminal of the primary coil **112** and the ground terminal GND. The clamp switch **140** has a first terminal and a second terminal, the first terminal of the clamp switch **140** may be coupled to the second terminal of the clamp capacitor **120**, and the second terminal of the clamp switch **140** may be coupled to the first terminal of the main switch **130**. That is, the clamp switch **140** may be connected in series between the clamp capacitor **120** and the main switch **130**, and a second terminal of the clamp switch **140** may also be coupled to the second terminal of the primary coil **112** such that the second terminal of the clamp switch **140**, the first terminal of the main switch **130**, and the second terminal of the primary coil **112** intersect at a common node N**1**.

[0023] The power converter **100** may be primarily operated in two stages. In the first stage, the control circuit **150** may turn on the main switch **130**, at which time the primary coil **112** may receive input voltage V.sub.in and store energy. Then, in the second stage, the control circuit **150** may turn off the main switch **130**, at which time the energy stored in the primary coil **112** is transferred to the secondary coil **114** by electromagnetic induction, and energy is transferred to the output terminal via the secondary coil **114** to maintain the output voltage V.sub.out. In addition, in the second stage, the power converter **100** may also store the leakage inductance energy of the primary coil **112** through the clamp capacitor **120**, enabling the power converter **100** to have a high power conversion efficiency. By repeatedly performing the operations of the first stage and the second stage, the power converter **100** can stably convert the input voltage V.sub.in into the output voltage V.sub.out and output same.

[0024] However, when the main switch **130** is turned on, if the node voltage VSW of the common node N**1** is relatively high, the main switch **130** will be turned on when the drain-source voltage is relatively high, and at this time, the main switch **130** will generate a relatively large switching loss, which is equivalent to enabling the main switch **130** to be turned on in a hard switching state. In order to solve this problem, in this embodiment, the control circuit **150** may turn on the clamp switch **140** for a period and then turn off before turning on the main switch **130** each time to pull down the node voltage VSW of the common node N**1**, thereby reducing the drain-source voltage of the main switch **130** and reducing a switching loss of the main switch **130**, which is equivalent to enabling the main switch **130** to be turned on in a soft switching state. In addition, in order to enable the drain voltage of the main switch **130** to be pulled down to a predetermined range to make the main switch **130** be turned on in the soft switching state as much as possible, in this embodiment, the control circuit **150** may detect whether the node voltage VSW of the common node N**1** is sufficiently low before turning on the main switch **130** each time, and determine the length of time for which the clamp switch **140** is turned on next time according to the embodiment.

[0025] As shown in FIG. **1**, the control circuit **150** may include a voltage detector **152** and a control signal generator **154**. The voltage detector **152** may compare node voltage VSW of common node N**1** with threshold voltage VTH**1**. In this embodiment, VTH**1** is set to 10 volts, however, a person skilled in the art may change its value as desired without being limited by this embodiment. The control signal generator **154** may generate a main control signal SIG.sub.M1 to control the main switch **130** and a clamp control signal SIG.sub.C1 to control the clamp switch **140**. The control signal generator **154** may periodically turn on the main switch **130** so that the power converter **100** can alternately and repeatedly perform the first and second stages of operation.

[0026] FIG. **2** is a flowchart of a method for operating M**1** of the power converter **100** according to an embodiment of the present disclosure; FIG. **3** is an operation timing diagram of the power converter **100** according to the method M**1**. As shown in FIG. **2**, the method M**1** may include repeating the steps S**110** to S**150**. In this embodiment, the main switch **130** may be, for example, an

N-type transistor, so in FIG. **3**, when the main control signal SIG.sub.M1 is at a logic high potential, the main switch **130** will be turned on, and the power converter **100** operates in the first stage; when the main control signal SIG.sub.M1 is at a logic low level, the main switch **130** is turned off and the power converter **100** operates in the second stage. In addition, the clamp switch **140** may be, for example, an N-type transistor, such that when the clamp control signal SIG.sub.C1 is at a logic high potential, the clamp switch **140** is turned on, and when the clamp control signal SIG.sub.C1 is at a logic low potential, the clamp switch **140** is turned off.

[0027] In this case, as shown in FIG. **3**, the clamp switch **140** is turned on during the periods TB**1**, TB**2** and TB**3**, the clamp switch **140** is turned off after the periods TB**1**, TB**2** and TB**3**, and the main switch **130** is turned on during the periods TA**1**, TA**2** and TA**3**. Further, the voltage detector **152** detects the magnitude relationship between the node voltage VSW of the common node N**1** and the threshold voltage VTH**1** at predetermined time points TC**1**, TC**2** and TC**3** before the periods TA**1**, TA**2** and TA**3**, and determines the length of time for which the next clamp switch **140** is turned on.

[0028] For case of understanding, the following description may refer to FIGS. **1**, **2**, and **3** simultaneously. In this embodiment, the control circuit **150** may turn on the clamp switch **140** (step S**110**) before each time the main switch **130** is turned on to generate a reverse current into the second terminal of the primary coil **112**. After the clamp switch **140** is turned on for a period, the control circuit **150** turns off the clamp switch **140** before turning on the main switch **130** (step S**120**), at which time the reverse current draws a charge from the equivalent capacitor C.sub.p on the common node N**1**, i.e., discharges the equivalent capacitor C.sub.p, thereby pulling down the voltage of the node voltage VSW of the common node N**1**. As a result, the main switch **130** can be turned on with a small drain-source voltage (step S**130**), thereby reducing a switching loss of the main switch **130**.

[0029] Since the equivalent capacitor C.sub.p on the common node N**1** may include the overall equivalent capacitor of a plurality of capacitors, such as the parasitic capacitor of the main switch **130**, the parasitic capacitor of the clamp switch **140** and the mirror capacitor at the corresponding position on the secondary side, the magnitude of the equivalent capacitor C.sub.p may be related to the manufacturing process, the circuit where it is located and the operating environment (e.g. temperature). In this case, even if the power converter **100** is generated based on the same design, the equivalent capacitor C.sub.p on its common node N**1** may be different in actual operation. That is, each of the power converters **100** must discharge the equivalent capacitor C.sub.p differently in order to pull down the node voltage VSW to the appropriate range.

[0030] In order to enable the main switches in different power converters **100** to be turned on in the soft switching state; in this embodiment, the control signal generator **154** may determine the magnitude relationship between the node voltage VSW and the threshold voltage VTH**1** detected by the voltage detector **152** at a predetermined time point after the clamp switch **140** is turned off and before the main switch **130** is turned on (step S**140**), and determine a length of time that the clamp switch **140** is turned on before the main switch **130** is turned on next time (step S**150**). Thus, the voltage of the common node N**1** can be pulled down by passing an appropriate discharge current before the main switch **130** is turned on next time, thereby more effectively suppressing a switching loss of the main switch **130**.

[0031] For example, as shown in FIG. **3**, the node voltage VSW detected by the voltage detector **152** at the time point TC**1** is greater than the threshold voltage VTH**1** before the time period TA**1** when the main switch **130** is turned on, indicating that the drain-source voltage of the main switch **130** is still high when the main switch is turned on at the time period TA**1**, so that the control signal generator **154** may extend the length of time for which the clamp switch **140** is turned on before the main switch **130** is turned on next time (i.e., before the time period TA**2**) to increase the backflow into the second terminal of the primary coil **112**. That is, the length of the time period TB**2** during which the clamp switch **140** is turned on for the second time is greater than that of the time period

TB**1** during which the clamp switch **140** is turned on for the first time, so that the clamp switch **140** will be turned on longer in the time period TB**2** and generate a larger reverse current so that the power converter **100** can discharge the equivalent capacitor C.sub.p by the larger reverse current during the time period after the clamp switch **140** is turned off and before the main switch **130** is turned on (i.e., the time period between the time periods TB**2** and TA**2**), thereby pulling the node voltage VSW further down to a lower level before the main switch **130** is turned on. As shown in FIG. **3**, the node voltage VSW detected by the voltage detector **152** at a time point TC**2** is smaller than the node voltage VSW detected by the voltage detector **152** at a time point TC**1**, so that the switching loss when the main switch **130** is turned on again at the time period TA**2** is low.
[0032] Further, in this embodiment, when the voltage detector **152** detects that the node voltage VSW is less than the threshold voltage VTH**1** at the predetermined time point TC**2**, the control signal generator **154** maintains and fixes the length of time for which the clamp switch **140** is turned on before the main switch **130** is turned on subsequently each time. For example, in FIG. **3**, since the node voltage VSW detected by the voltage detector **152** at the time point TC**2** is less than the threshold voltage VTH**1**, the control signal generator **154** may cause the clamp switch **140** to be turned on next time for a same length of time, i.e., the length of the time period TB**3** may be equal to that of the time period TB**2**.
[0033] However, this application is not limited thereto. In some embodiments, the operating environment of the power converter **100**, such as temperature, may change over time, and the input voltage Vin may also change over time in a test environment; in this case, the time that the fixed clamp switch **140** is turned on may not effectively suppress a switching loss of the main switch **130** for a long time. Thus, the control signal generator **154** may also adjust the time that the clamp switch **140** is turned on according to other mechanisms, not necessarily making the time that subsequent clamp switches **140** are turned on be fixed.
[0034] FIG. **4** is an operation timing diagram of the power converter **100** according to another embodiment. In FIG. **4**, when the voltage detector **152** detects that the node voltage VSW is greater than the threshold voltage VTH**1** at the time point TC**1**, the control signal generator **154** may extend the clamp switch **140** is turned on next time, such that the length of the time period TB**2** is greater than that of the time period TB**1**. However, when the voltage detector **152** detects that the node voltage VSW is less than the threshold voltage VTH**1** at the time point TC**2**, the control signal generator **154** may shorten the clamp switch **140** is turned on next time, such that the length of the time period TB**3** is less than that of the time period TB**2**. As such, the control signal generator **154** may also dynamically adjust the time that the clamp switch **140** is turned on as operating conditions change.
[0035] FIG. **5** is an operation timing diagram of the power converter **100** according to another embodiment. In the embodiment of FIG. **5**, the voltage detector **152** may compare the node voltage VSW with two threshold voltages VTH**1** and VTH**2** at predetermined time points TC**1**, TC**2**, TC**3** and TC**4**, and adjust the length of time for which the next clamp switch **140** is turned on according to the magnitude relationship between the node voltage VSW and the threshold voltages VTH**1** and VTH**2**, where the threshold voltage VTH**2** is less than the threshold voltage VTH**1**.
[0036] For example, when the voltage detector **152** detects that node voltage VSW is greater than threshold voltage VTH**1** at the time point TC**1**, the control signal generator **154** may extend the length of time for which the clamp switch **140** is turned on next time, such that the length of the time period TB**2** is greater than that of the time period TB**1**. When voltage detector **152** detects that node voltage VSW is less than threshold voltage VTH**1** and greater than threshold voltage VTH**2** at the time point TC**2**, control signal generator **154** may maintain the length of time for which the clamp switch **140** is turned on next time, such that the length of the time period TB**3** is equal to that of the time period TB**2**. However, if the voltage detector **152** detects that node voltage VSW is less than threshold voltage VTH**2** at the time point TC**3**, the control signal generator **154** may shorten the length of time for which the clamp switch **140** is turned on next time, such that the length of the

time period TB**4** is less than that of the time period TB**3**. In this embodiment, VTH**1** is set to 10 volts and VTH**2** is set to 5 volts, however a person skilled in the art may change their values as desired without being limited by this embodiment.

[0037] That is, the control signal generator **154** adjusts the length of time for which the clamp switch **140** is turned on next time only when the node voltage VSW is greater than the threshold voltage VTH**1** or less than the threshold voltage VTH**2**, and maintains the length of time for which the clamp switch **140** is turned on next time when the node voltage VSW is between the threshold voltage VTH**1** and the threshold voltage VTH**2**. As such, it is possible to avoid adjusting the time of turning on the clamp switch **140** too frequently, thereby achieving a hysteresis-like control effect. In other embodiments, the power converter **100** may be provided with more than two threshold voltages as desired.

[0038] Further, in the embodiment of FIG. **1**, the voltage detector **152** may include a comparator **1521**. A first terminal of the comparator **1521** may be coupled to a first terminal of the main switch **130**, and a second terminal of the comparator **1521** may receive the threshold voltage VTH**1**. That is, the voltage detector **152** may directly compare the node voltage VSW of the common node N**1** with the threshold voltage VTH**1** through the comparator **1521**. However, the present disclosure is not limited thereto. In some embodiments, directly using node voltage VSW as the input voltage to the comparator **1521** may result in damage to the comparator **1521** because node voltage VSW may have a higher voltage value. In this case, the voltage detector **152** may compare the magnitude relationship between the node voltage VSW and the threshold voltage VTH**1** in other manners.

[0039] FIG. **6** is a schematic diagram of a power converter according to another embodiment of the present disclosure. The power converter **200** may have a similar structure and operate according to similar principles as the power converter **100**, however, the voltage detector **252** in the power converter **200** may include resistors RA**1**, RA**2** and comparator **2521**. A first terminal of the resistor RA**1** may be coupled to a first terminal of a main switch **130**, a first terminal of the resistor RA**2** may be coupled to a second terminal of the resistor RA**1**, and a second terminal of the resistor RA**2** may be coupled to a ground terminal GND. Further, a first terminal of the comparator **2521** may be coupled to a second terminal of the resistor RA**1**, and a second terminal of the comparator **2521** may receive a reference voltage VRA**1** corresponding to the threshold voltage VTH**1**.

[0040] That is, the voltage detector **252** may divide node voltage VSW through resistors RA**1** and RA**2** to reduce the voltage received at the first terminal of the comparator **2521**. In this case, the reference voltage VRA**1** may also be less than the threshold voltage VTH**1**, and the ratio of the reference voltage VRA**1** to the threshold voltage VTH**1** may be equal to that of the resistor RA**2** to the sum of the resistances RA**1** and RA**2**, i.e., RA**2**/(RA**1**+RA**2**). As such, the voltage detector **252** may be implemented using a conventional low withstand voltage comparator **2521**.

[0041] FIG. **7** is a schematic diagram of a power converter according to another embodiment of the present disclosure. The power converter **300** may have a similar structure and operate according to similar principles as a power converter **100**, however, the voltage detector **352** in the power converter **300** may include a diode D**1**, a resistor RB**1**, and a comparator **3521**. A cathode of the diode D**1** may be coupled to a first terminal of the main switch **130**, a first terminal of the resistor RB**1** may be coupled to an anode of diode D**1**, and a second terminal of the resistor RB**1** may receive a predetermined bias voltage VD**1**. A first terminal of the comparator **3521** may be coupled to a first terminal of the resistor RB**1**, and a second terminal of the comparator **3521** may receive a reference voltage VRB**1** corresponding to the threshold voltage VTH**1**.

[0042] In this embodiment, the predetermined bias voltage VD**1** may be greater than the reference voltage VRB**1**. In this case, when the node voltage VSW is greater than the predetermined bias voltage VD**1**, the diode D**1** will be in a reverse bias state, so that the first terminal of the comparator **3521** will receive the predetermined bias voltage VD**1**, at which time the comparator **3521** will output the first voltage. Conversely, when the node voltage VSW is less than the predetermined bias voltage VD**1** minus the threshold voltage of the diode D**1**, the diode D**1** will be in a forward bias state, so that the first terminal of the comparator **3521** will receive the node voltage VSW plus the threshold voltage of the diode D**1**, at which time the comparator **3521** will output a second voltage different from the first voltage. That is, by appropriately setting the predetermined bias voltage VD**1** and the reference voltage VRB**1**, it is possible for the comparator **3521** to compare the node voltage VSW with the reference voltage VRB**1** only when the node voltage VSW is small, thereby preventing the comparator **3521** from receiving an excessively high voltage.

[0043] FIG. **8** is a schematic diagram of a power converter according to another embodiment of the present disclosure. The power converter **400** may have a similar structure and may operate according to similar principles as the power converter **100**, however, the power converter **400** may include two primary coils **112** and **412** and the voltage detector **452** may include resistors RC**1**, RC**2** and the comparator **4521**. In this embodiment, the power converter **400** may generate the sensing voltage VS**1** corresponding to the node voltage VSW through the primary coil **412**, and may divide the sensing voltage VS**1** through the resistors RC**1** and RC**2**.

[0044] As shown in FIG. **8**, a first terminal of the resistor RC**1** may be coupled to a first terminal of the primary coil **412**, a first terminal of the resistor RC**2** may be coupled to a second terminal of the resistor RC**1**, and a second terminal of the resistor RC**2** may be coupled to a second terminal of the primary coil **412** and a ground terminal GND. Further, a first terminal of the comparator **4521** may be coupled to a second terminal of the resistor RC**1**, and a second terminal of the comparator **4521** may receive reference voltage VRC**1** corresponding to threshold voltage VTH**1**.

[0045] In this embodiment, since the power converter **400** generates the sensing voltage VS**1** corresponding to the node voltage VSW through the primary coil **412** and divides the voltage through the resistors RC**1**, RC**2**, the influence on the node voltage VSW in the detection process can be reduced, and the input voltage of the comparator **4521** can be reduced.

[0046] Further, in the power converters **100**, **200**, **300** and **400**, the control signal generator **154** thereof may control the clamp switch **140** by generating the clamp control signal SIG.sub.C1 in addition to the periodic square wave signal SIG.sub.M1 to control the main switch **130**, and may adjust the time for which the clamp switch **140** is turned on each time by adjusting the pulse width of the clamp control signal **140**. For example, in the embodiment of FIG. **1**, the control circuit **150** may include an analog circuit and the control signal generator **154** may include a comparator (not shown) that may compare the reference voltage VR**0** with a sawtooth wave SIGs (or a triangular wave). In this case, the control signal generator **154** may control the pulse width of the clamp control signal SIG.sub.C1 by adjusting the move mode of the sawtooth wave SIGs (or a triangular wave) or adjusting the level of the reference voltage VR**0**.

[0047] However, the present disclosure is not limited thereto, and in some other embodiments, the control circuit **150** may include a digital circuit and the control signal generator **154** may include a register and a counter. In this case, the control signal generator **154** may store a value related to the pulse width in the register, and the counter may count the clock signal in the digital circuit to control the pulse width of the clamp control signal SIG.sub.C1 according to the value stored in the register.

[0048] Further, in some embodiments, the control circuit **150** may be provided, for example, in a separate chip CP**1**, so that the circuit designer may more conveniently use the chip CP**1** to detect the voltage of the primary coil therein and control the clamp switch thereof and the main switch to be turned on in various power converter designs. Similarly, the control circuits **250**, **350**, and **450** may also be designed as chips for use by circuit designers.

[0049] In summary, embodiments of the present disclosure provide a power converter, a semiconductor chip, and a method for operating a power converter. Since the length of time for which the clamp switch is turned on next time can be determined by the power converter of the present disclosure according to the amplitude of the node voltage of the common node before the main switch is turned on each time, the voltage of the common node can be pulled down to a predetermined range in different circuits and environments, thereby effectively suppressing a switching loss of the main switch.

[0050] Spatially relative terms, such as “below”, “under”, “lower”, “above”, “upper”, “left”, “right”, and the like, may be used herein for ease of description to describe relationship between one component or feature and another component(s) or feature(s) as illustrated in the figures. Spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The devices may be otherwise oriented (rotated **90** degrees or at other orientations) and the spatially relative descriptors used herein may be interpreted in a corresponding manner. It will be understood that when an element is referred to as being “connected” or “coupled” to another element, it can be directly connected or coupled to the other element or intervening elements may be present.

[0051] As used herein, the terms “about”, “substantially”, “generally”, and “approximately” are used to describe and explain minor variations. When used in connection with an event or circumstance, the term can refer to instances where the event or circumstance occurs precisely as well as instances where the event or circumstance occurs nearly. As used herein with respect to a given value or range, the term “about” generally means within ±10%, ±5%, ±1%, or ±0.5% of the given value or range. Ranges can be expressed herein as from one endpoint to another endpoint, or between two endpoints. All ranges disclosed herein are inclusive of the endpoints, unless otherwise specified. The term “substantially coplanar” can mean that two surfaces are positioned within a few micrometers (μm) of difference in position along the same plane, such as within 10 μm, within 5 μm, within 1 μm, or within 0.5 μm of difference in position along the same plane. When numerical values or characteristics are referred to as being “substantially” the same, the term can refer to values that are within ±10%, ±5%, ±1%, or ±0.5% of the average of the stated values.

[0052] The foregoing has outlined features of several embodiments and detailed aspects of the disclosure. The embodiments described in the present disclosure may be readily utilized as a basis for designing or modifying other processes and structures for carrying out the same or similar purposes and/or to achieve the same or similar advantages of the embodiments described herein. Such equivalent constructions do not depart from the spirit and scope of the present disclosure, and various changes, substitutions, and alterations may be made without departing from the spirit and scope of the present disclosure.

Claims

1. A power converter comprising: a transformer having a first primary coil and a secondary coil, the first primary coil being configured to receive an input voltage and the secondary coil being configured to generate an output voltage; a clamp capacitor having a first terminal coupled to a first terminal of the first primary coil; a main switch coupled to a second terminal of the first primary coil; a clamp switch coupled in series between a second terminal of the clamp capacitor and the main switch, wherein the main switch, the clamp switch, and the first primary coil intersect at a node; and a control circuit configured to: periodically turn on the main switch to charge the first primary coil; turn on the clamp switch before the main switch to generate a reverse current through the first primary coil; turn off the clamp switch before the main switch is turned on to discharge an equivalent capacitor on the node coupled to the main switch; and adjust an operation of the clamp switch based on a monitored voltage at the node, thereby reducing switching losses of the main switch.

2. The power converter of claim 1, wherein the control circuit is configured to compare the monitored voltage at the node to a first threshold voltage to adjust a length of time for which the clamp switch is turned on.

3. The power converter of claim 2, wherein when the monitored voltage is greater than the first threshold voltage, the control circuit extends the length of time for which the clamp switch is

turned on before the main switch is turned on a next time.
4. The power converter of claim 2, wherein when the monitored voltage is less than the first threshold voltage, the control circuit maintains the length of time for which the clamp switch is turned on before the main switch is turned on subsequently each time.
5. The power converter of claim 2, wherein when the monitored voltage is less than the first threshold voltage, the control circuit reduces the length of the time for which the clamp switch is turned on before the main switch is turned on a next time.
6. The power converter of claim 2, wherein when the monitored voltage is less than the first threshold voltage and greater than a second threshold voltage, the control circuit maintains the length of time for which the clamp switch is turned on before the main switch is turned on, wherein the second threshold voltage is less than the first threshold voltage.
7 The power converter of claim 6, wherein when the monitored voltage is less than the second threshold voltage at a predetermined time point, the control circuit reduces the length of time for which the clamp switch is turned on before the main switch is turned on a next time.
8. The power converter of claim 1, further comprising a voltage detector coupled to the node and configured to detect the monitored voltage at the node.
9 The power converter of claim 8, wherein the voltage detector comprises a first resistor and a second resistor connected in series.
10. The power converter of claim 8, wherein the voltage detector comprises a diode, a resistor, and a comparator.
11. The power converter of claim 1, further comprising a second primary coil configured to generate a sensing voltage corresponding to a voltage at the node.
12. The power converter of claim 11, wherein a voltage detected by the second primary coil is divided by resistors and input to the control circuit.
13. The power converter of claim 1, wherein the equivalent capacitor on the node comprises a parasitic capacitance associated with the main switch.
14. The power converter of claim 1, wherein the transformer comprises a flyback transformer.
15. The power converter of claim 1, wherein the main switch and the clamp switch are MOSFETs.
16. The power converter of claim 15, wherein the control circuit is configured to generate gate control signals for turning on and off the main switch and the clamp switch.
17. The power converter of claim 1, wherein the equivalent capacitor is related to a temperature of the power converter.
18. The power converter of claim 1, further comprising a bias voltage circuit configured to provide a reference voltage to the control circuit for comparison with the monitored voltage at the node.
19. The power converter of claim 1, wherein the control circuit comprises a digital circuit including a counter for adjusting a length of time for which the clamp switch operated based on clock signals.
20. The power converter of claim 1, wherein the control circuit is configured to turn off the clamp switch after a predetermined time delay before turning on the main switch.